

Comparative pathway enrichment across histone-mark peak sets in C1, C2, T1, and T2

Across the four samples (C1, C2, T1, T2), pathway enrichments are sharply concentrated within a small subset of histone-mark pairs, with broadly weak signals elsewhere. A common axis of endocrine responsiveness is evident: **ESTROGEN_RESPONSE_EARLY/LATE** is strongly enriched in C1 and T1 and recurs in C2 and T2 (Figures 1–4), consistent with robust estrogen-regulated transcriptional modules encoded by specific chromatin states (Lee et al., 2022). **KRAS_SIGNALING_DN** is repeatedly detected (C1, C2, T1) and appears as an isolated high signal in a single T2 column, while **P53_PATHWAY** is prominent in C1, T1, and T2 (Figures 1, 3, 4). **DNA_REPAIR** shows moderate-to-strong enrichment in C1, C2, and T2. Proliferation-linked modules (**E2F_TARGETS**, **MYC_TARGETS_V1**, **G2M_CHECKPOINT**) are most coherent in T2, weaker and scattered in C2, and limited to moderate **MYC_TARGETS_V1** in T1 (Figures 2–4), aligning with canonical E2F/MYC-centered cell-cycle circuitry (Lin et al., 2023). Epithelial polarity programs (**APICAL_SURFACE**, **APICAL_JUNCTION**) are most evident in C1–C2, with **APICAL_SURFACE** also appearing as a single-column peak in T2 (Figures 1, 2, 4). In contrast, immune and metabolic gene sets are consistently attenuated: interferon responses and inflammatory modules are faint in C1 and T1; **COMPLEMENT**, **COAGULATION**, and **BILE_ACID_METABOLISM** are weak in T2; and **GLYCOLYSIS** and **OXIDATIVE_PHOSPHORYLATION** are uniformly low in C1–C2 (Figures 1–4).

These biological themes track with mark specificity. In C2, the strongest signals cluster in H3K27ac-anchored pairs (e.g., H3K27ac–H3K4me1), particularly for **ESTROGEN_RESPONSE_LATE**, **APICAL_SURFACE**, **KRAS_SIGNALING_DN**, and **DNA_REPAIR**, whereas rightmost columns that include the bivalent H3K27me3–H3K4me3 pair show minimal enrichment (Figure 2). In T2, multiple pathways (**P53_PATHWAY**, **MYC_TARGETS_V1**, **G2M_CHECKPOINT**, **E2F_TARGETS**, **ANGIOGENESIS**, **DNA_REPAIR**, and both **ESTROGEN_RESPONSE** sets) are enriched across several columns, but the H3K27me3–H3K9me3 column exhibits little signal (Figure 4). C1 manifests its highest enrichment cluster toward the rightmost pairs, led by the two **ESTROGEN_RESPONSE** sets and accompanied by **KRAS_SIGNALING_DN**, **APICAL_SURFACE**, and **P53_PATHWAY** (Figure 1). T1 shows a single central group concentrating **ESTROGEN_RESPONSE_EARLY/LATE** with coincident **KRAS_SIGNALING_DN** and **P53_PATHWAY**; **MYC_TARGETS_V1** is only moderate and focal (Figure 3). Together, these patterns indicate that active enhancer–promoter chromatin—most notably H3K27ac with H3K4me1—preferentially encodes endocrine,

epithelial polarity, DNA damage/p53, and proliferation programs, whereas repressive/bivalent pairs (e.g., H3K27me3–H3K4me3; H3K27me3–H3K9me3) contribute little to pathway overrepresentation in this setting.

In summary, chromatin-associated enrichments converge on shared endocrine and stress/proliferation axes with sample-specific emphases: C2 favors H3K27ac-dominated enhancer states engaging estrogen and epithelial polarity programs; T2 uniquely consolidates proliferation, cell-cycle, and angiogenesis signals across multiple pairs; T1 concentrates hormone and p53 pathways within a single active group; and C1 mirrors the endocrine–epithelial polarity–p53 triad with polarized rightmost columns. Immune and broad metabolic pathways are comparatively depleted across most mark-defined peak sets. These

comparative patterns underscore conserved regulatory themes alongside mark-dependent, sample-specific specialization in pathway engagement (Lee et al., 2022; Lin et al., 2023).

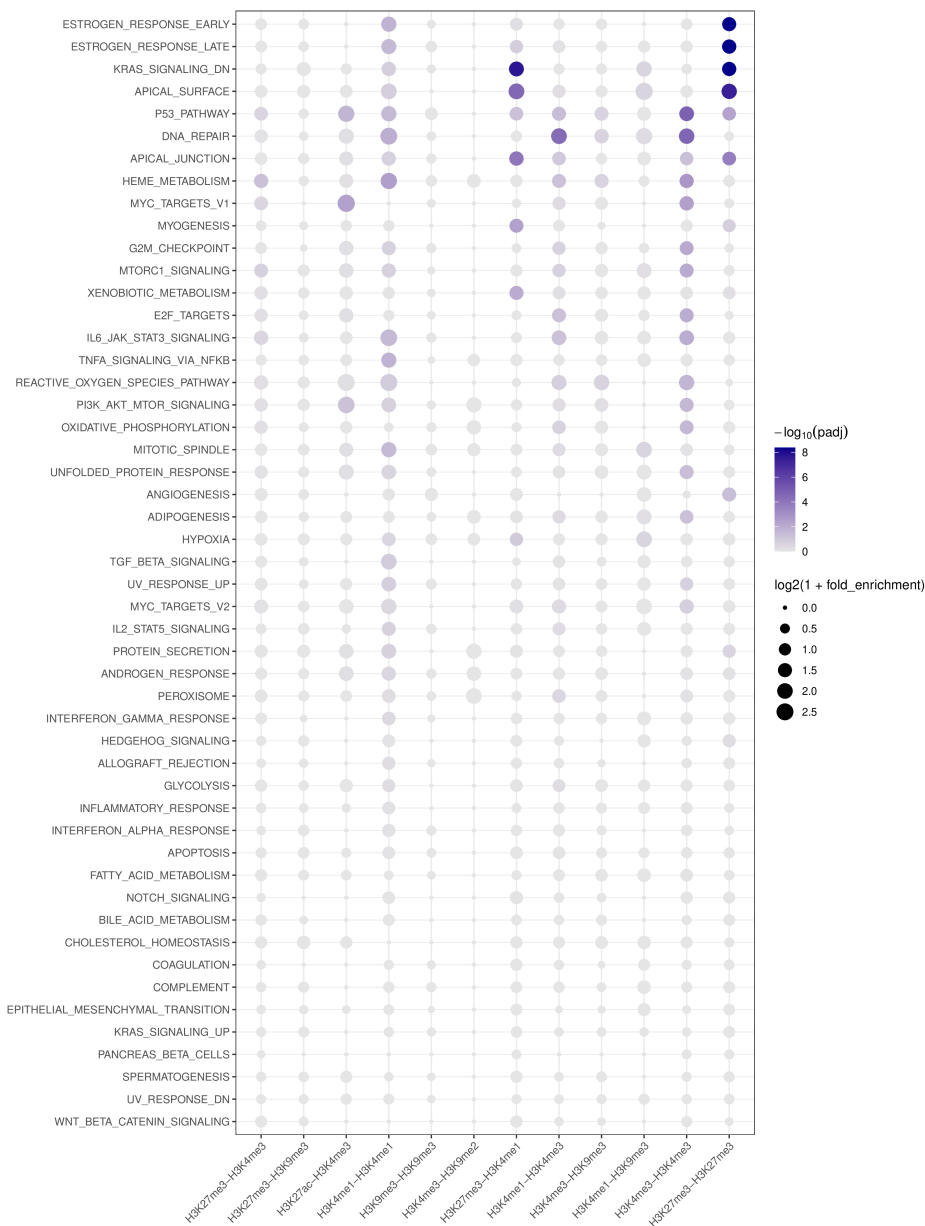


Figure 1. Pathway enrichment across histone-mark peak sets in C1. Bubble plot of hallmark pathway enrichment for genes linked to C1 peak sets across histone-mark combinations. Strong signals concentrate in estrogen response (early/late), KRAS signaling down, apical surface, and p53 pathway within a subset of combinations, while most other pathways show minimal enrichment. Bubble color reflects adjusted P value and size represents fold-enrichment.



Figure 2. Enriched pathways across histone peak combinations in C2. Pathway enrichment is summarized for peaks grouped by histone-mark pairs. Larger and darker bubbles indicate higher fold enrichment and significance. Signals concentrate in APICAL_SURFACE, KRAS_SIGNALING_DN, and ESTROGEN_RESPONSE_LATE for H3K27ac-linked groups, with most other combinations showing little enrichment.



Figure 3. Pathway enrichment of genes linked to T1 peaks. Enrichment signals are displayed across histone mark-defined groups for pathways associated with T1 peak-linked genes. The strongest signals cluster in a single group for *ESTROGEN_RESPONSE_EARLY/LATE*, with *KRAS_SIGNALING_DN* and *P53_PATHWAY* also elevated, whereas most other pathways are weak or diffuse.

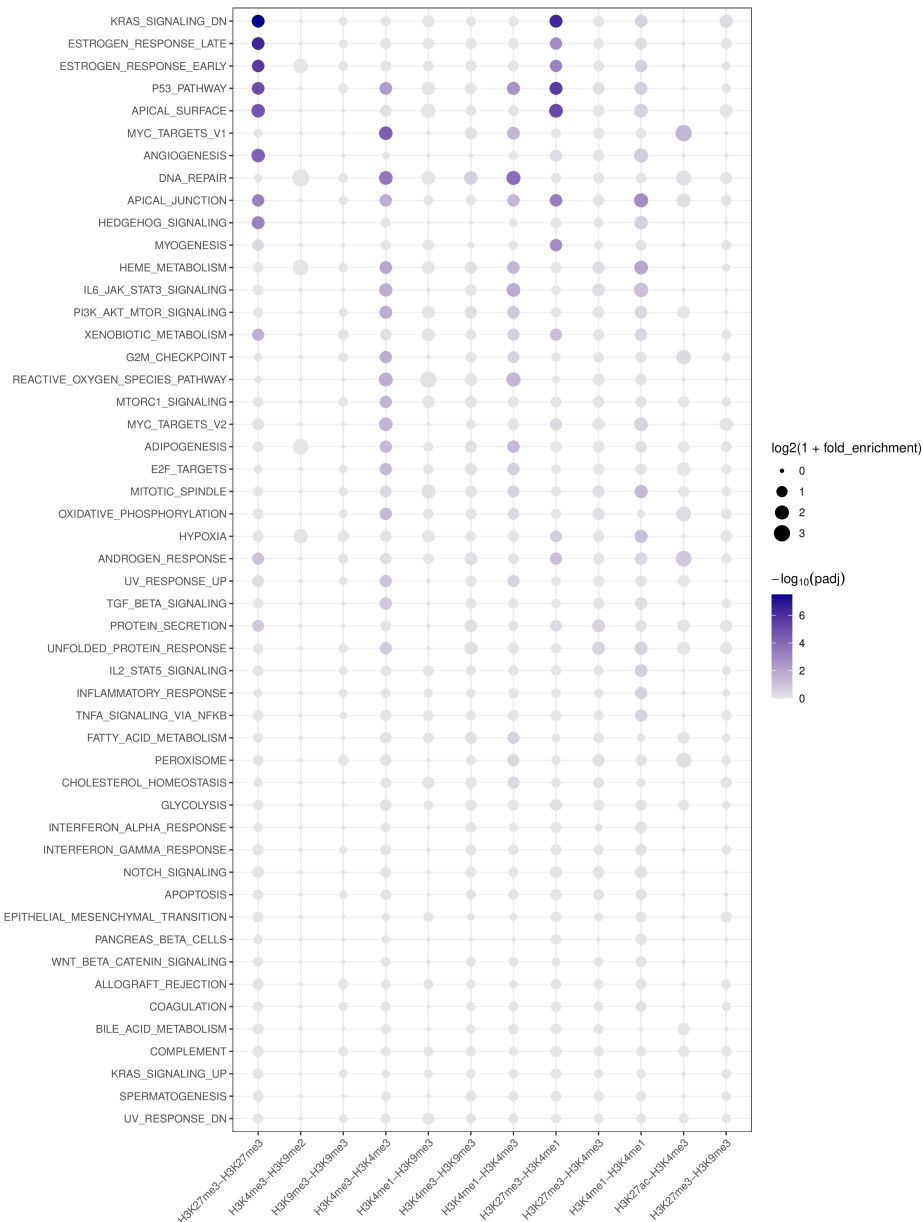


Figure 4. Pathway enrichment across histone-mark combinations in T2. Bubble plot of Hallmark pathways linked to genes near T2 peaks across different histone-mark combinations. Enrichment concentrates in pathways such as P53_PATHWAY, MYC targets, G2M_CHECKPOINT, and ANGIOGENESIS, with sparse signals for many immune and metabolic sets; the H3K27me3+H3K9me3 column shows little enrichment.

Transcription factor motif programs associated with C2 and T2 peaks

We compared transcription factor binding site (TFBS) motif enrichment across C2 and T2 peak sets to delineate shared versus condition-skewed regulatory programs (Figures 5–6). Two broad features emerged. First, both heatmaps contain clustered motif blocks with concordant enrichment across multiple motif resources, indicating conserved regulatory modules. Second, a subset of columns exhibits a global attenuation of scores forming a vertical depletion band that is largely motif-agnostic, superimposed on otherwise positive

landscapes; this band is prominent in C2 and detectable as a left-side low-value stripe in T2.

In C2 (Figure 5), several motif clusters show consistent moderate-to-strong enrichment across adjacent columns, pointing to stable modules preserved despite library-to-library variation. Alongside these conserved blocks, we observed alternating high/low tracts and localized patches, indicative of column-sensitive modules whose apparent strength depends on the motif resource or implementation. The pronounced depletion band spans nearly all rows with few neutral exceptions, arguing for a column-specific effect (e.g., library or scoring-scale attenuation) rather than selective depletion of particular TF families. Thus, C2 appears to harbor a conserved backbone of regulatory inputs overlaid by column-dependent heterogeneity.

In T2 (Figure 6), enrichment is globally stronger, with most rows positive and fewer depleted tiles, and with broad concordance between JASPAR and HOCOMOCO columns except for occasional departures. Multiple TF families form sustained, cross-library blocks—most notably SOX, KLF, ETS, TEAD, and GATA—consistent with a cohesive program of lineage and chromatin-activation factors. In contrast, subsets labeled as ZNF and IRF display mixed or weaker signals, implying either secondary or context-limited engagement relative to the dominant program. The left-side stripe of lower values, similar in form to the C2 depletion band, again involves many rows, reinforcing a column-specific attenuation rather than a family-specific loss. Scattered blue tiles punctuating otherwise red rows underscore modest library-specific differences atop a robust, conserved signal.

Taken together, these patterns support the following inferences: - Conserved modules: Both C2 and T2 contain motif clusters with stable, cross-library enrichment, providing high-confidence evidence for shared regulatory inputs. In T2, these conserved blocks prominently comprise SOX/TEAD/ETS/GATA/KLF families, aligning with known cell-identity and enhancer-activation circuits; the presence of KLF motifs is consistent with potential involvement of KLF factors such as KLF4 in chromatin regulation (An et al., 2024). - Sample-shifted modules: Column-dependent alternation in C2 and library-specific exceptions in T2 indicate modules whose apparent strength is sensitive to motif resource/implementation. These should be interpreted cautiously and prioritized below the conserved blocks for mechanistic attribution. - Depletion bands: The vertical attenuation affecting many motifs within specific columns (prominent in C2; a left-side stripe in T2) most parsimoniously reflects column-specific factors (e.g., motif-set composition, background modeling, or scoring scale) rather than biological silencing of broad TF

classes. Downstream interpretation should therefore emphasize consensus across the remaining concordant columns.

Overall, T2 exhibits a more unified motif program dominated by SOX, KLF, ETS, TEAD, and GATA families, whereas C2 shows a comparable conserved backbone with greater column-sensitive heterogeneity. Emphasizing the conserved, cross-library blocks and

down-weighting the column-wide depletion band provides a reliable motif-based blueprint of the regulatory modules that differentially shape the C2 versus T2 chromatin landscapes.

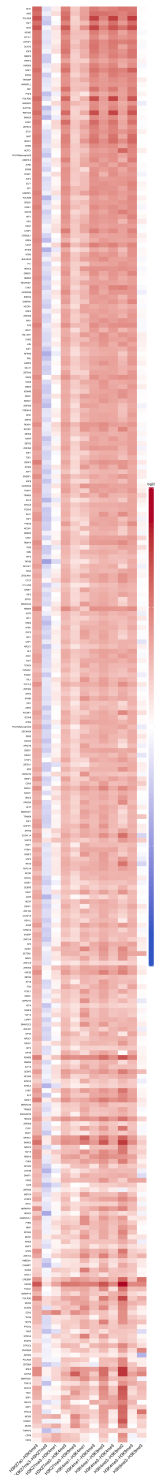


Figure 5. TFBS motif enrichment across C2 peaks. Heatmap summarizing motif enrichment for the C2 peak set. Most motifs are positively enriched across columns, while one column shows broad depletion; clustering reveals subsets with uniformly strong or mixed signals.

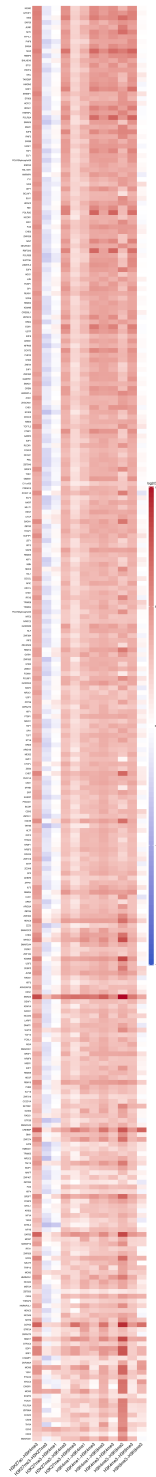


Figure 6. TFBS motif enrichment across databases for T2 peaks. *Enrichment scores for transcription factor binding motifs in the T2 peak set are shown across multiple motif libraries. Most motifs display positive enrichment with occasional weak or negative signals, and clustering groups motifs with similar patterns that point to candidate regulators in T2.*

Regulatory element class composition (cCRE) across histone-mark pairs in C2 and T2

Intersecting C2 and T2 peak sets with the ENCODE v4 candidate cis-regulatory element (cCRE) registry showed that regulatory-element class composition tracks with the underlying chromatin environment of each histone-mark pair (Figures 7–8) (The ENCODE Project Consortium et al., 2020). Across pairs, both C2 and T2 are globally promoter-skewed, with promoter-like signatures (PLS) contributing the largest fraction and enhancer-like signatures (pELS/dELS) comprising a secondary share; CA/TF-associated subclasses (CA–CTCF, CA–TF, CA–H3K4me3, TF) are generally smaller. Activating mark combinations reinforce these patterns: H3K4me3-containing pairs are consistently PLS-dominant in both C2 and T2 (e.g., H3K4me3–H3K4me3 and H3K27ac–H3K4me3), concordant with the promoter association of H3K4me3 and the activation-linked H3K27ac mark (Creyghton et al., 2010). By contrast, H3K4me1-containing pairs in C2 shift composition toward enhancers, showing elevated dELS and pELS relative to PLS, consistent with H3K4me1 marking poised/primed enhancers (Creyghton et al., 2010; Rada-Iglesias et al., 2011). In T2, enhancer contributions are also conspicuous for H3K27me3–H3K27me3 (increased dELS) and for H3K27me3 paired with H3K4me1 or H3K4me3 (noticeable pELS alongside PLS), aligning with mixed or bivalent promoter–enhancer contexts when Polycomb and activating marks coexist.

Repressive marks reshape cCRE composition away from promoter-like states. In C2, H3K9me3-bearing pairs—especially H3K9me3–H3K9me3—show a pronounced depletion of PLS and a concomitant rise in CA-associated subclasses and “other,” indicative of heterochromatin-rich contexts where classical promoter/enhancer annotations are underrepresented and accessible islands (including CTCF-associated sites) predominate (Becker et al., 2017) (Figure 7). T2 shows the same directionality for H3K9me3–H3K9me3, with comparatively elevated CA–H3K4me3, CA–CTCF, CA–TF, and TF categories, while the “other” class remains minor for most pairs (Figure 8). H3K27me3 combinations display intermediate, mixed profiles: several H3K27me3 groups in C2 contain both promoter- and enhancer-like contributions, and in T2, H3K27me3–H3K27me3 is notably dELS-enriched and H3K27me3 paired with H3K4me1/3 retains substantial PLS with visible pELS, consistent with bivalency and poised regulatory states. Together, these observations demonstrate that activating marks (H3K27ac, H3K4me1/3) bias peak overlap toward promoter- and enhancer-like cCREs, whereas repressive marks (H3K27me3, H3K9me3) shift composition toward enhancer-like or CA/TF-associated classes and away from PLS, with H3K9me3 pairs showing the strongest CA enrichment

consistent with heterochromatic compaction (Creyghton et al., 2010; Rada-Iglesias et al., 2011; Becker et al., 2017; The ENCODE Project Consortium et al., 2020).

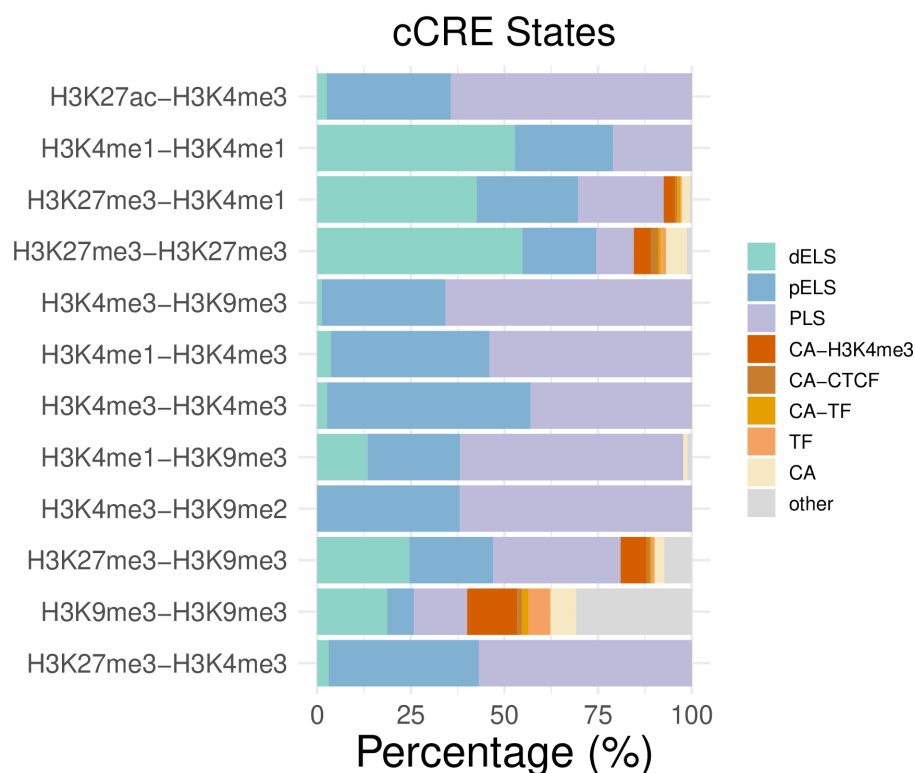


Figure 7. cCRE composition of C2 peaks across histone-mark pairs. C2 peak intervals were overlapped with cCRE annotations and summarized as fractions per histone-mark pair. Promoter-like states dominate most groups, whereas enhancer-like states are enriched for H3K4me1 groups and CA/other classes increase in H3K9me3 groups.

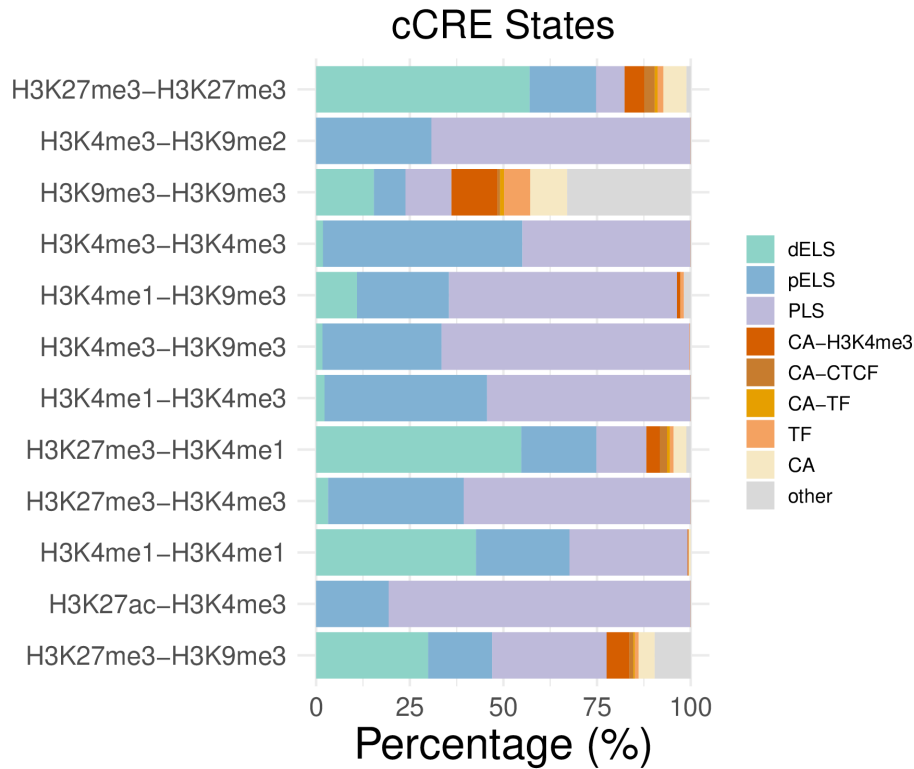


Figure 8. T2 peak composition across ENCODE cCRE classes. T2 peaks for each histone-mark combination are annotated to cCRE states. Promoter-like (PLS) comprises the largest share for most groups, with variable dELS/pELS contributions; CA- and TF-associated classes are generally small, with a relative increase for H3K9me3–H3K9me3 and a dELS enrichment for H3K27me3–H3K27me3.

Genic localization of peaks across histone-mark pairs in C2 and T2

We annotated peaks from each histone-mark pair to gene features (promoter, exon, intron, intergenic) and compared promoter-proximal versus intragenic/distal localization between C2 and T2. In C2, H3K4me3-containing pairs were consistently promoter-enriched, including H3K27ac–H3K4me3, H3K4me3–H3K4me3, H3K4me1–H3K4me3, H3K4me3–H3K9me2, H3K4me3–H3K9me3, and H3K27me3–H3K4me3 (Figure 9). H3K4me1–H3K4me1 retained a promoter bias but with a notable intronic share, whereas H3K27me3–H3K4me1 was more evenly split across exons and introns with a smaller promoter fraction. Pairs dominated by H3K27me3 or H3K9me3 alone (H3K27me3–H3K27me3, H3K27me3–H3K9me3, H3K9me3–H3K9me3) showed reduced promoter occupancy and increased intronic/intergenic assignments, with visible exon contributions (Figure 9). These patterns align with the canonical promoter-proximal localization of H3K4me3 and the broader intragenic/distal distribution of repressive marks (Samsudin et al., 2025).

In T2, peak localization was more polarized across pairs (Figure 10). Promoter fractions remained prominent for H3K4me3-containing pairs (H3K4me3–H3K4me3, H3K4me1–H3K4me3, H3K27ac–H3K4me3, H3K27me3–H3K4me3) and, notably, also for

H3K27me3–H3K9me3. By contrast, introns accounted for the largest share in H3K27me3–H3K27me3 and H3K27me3–H3K4me1, while the intergenic/other category was comparatively high for H3K9me3–H3K9me3. Exons and UTRs were generally minor across pairs, although 5' UTR segments were more visible for H3K4me3–H3K9me2 and H3K4me1–H3K4me1 (Figure 10).

Comparing samples, both C2 and T2 exhibit strong promoter enrichment for H3K4me3-containing pairs, consistent with the established association of H3K4me3 with active transcription start sites and, for H3K4me3 with H3K27me3, promoter-proximal bivalency (Samsudin et al., 2025). However, T2 displays greater separation between promoter-skewed and intragenic/distal-skewed groups. Relative to C2, T2 shows: (i) a more intron-dominant profile for H3K27me3–H3K4me1; (ii) a pronounced shift of H3K9me3–H3K9me3 toward intergenic/other; and (iii) unexpectedly high promoter representation for H3K27me3–H3K9me3. H3K4me1–H3K4me1 remains promoter-biased in both, with C2 showing a stronger intronic share and T2 highlighting 5' UTR contributions. Collectively, these results indicate a conserved promoter bias for H3K4me3-containing pairs across

samples, contrasted by sample-specific redistribution of H3K27me3- and H3K9me3-dominant pairs toward intronic and distal contexts.

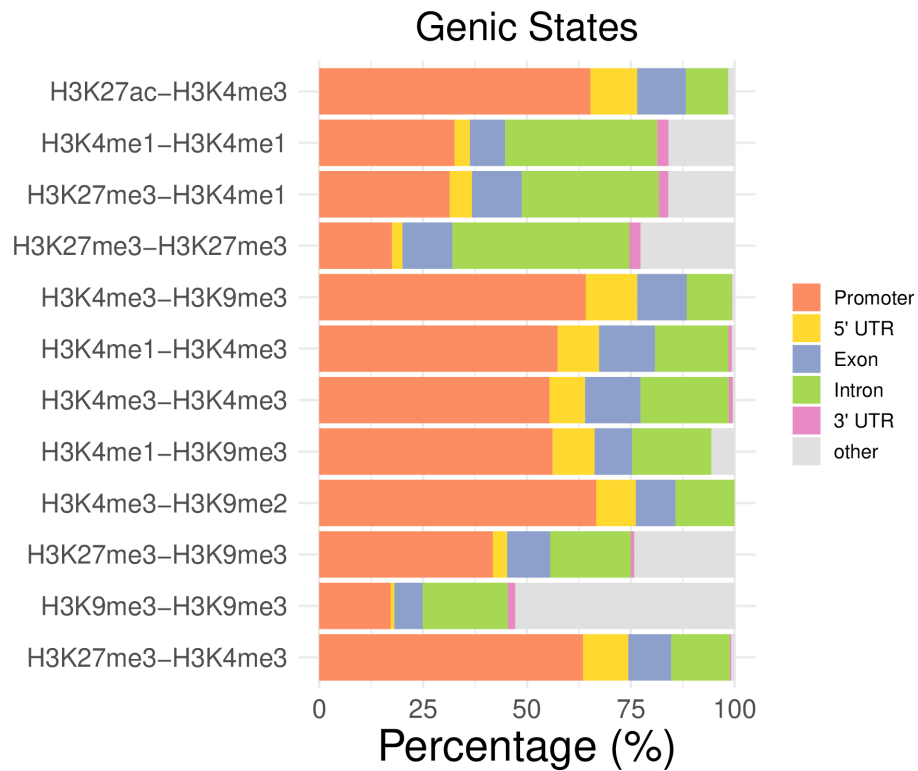


Figure 9. C2 peak genic distributions across histone mark pairs. Peaks were annotated to promoters, UTRs, exons, introns, or other regions. Mark pairs containing H3K4me3 (and H3K27ac) concentrate at promoters, whereas pairs dominated by H3K27me3 or H3K9me3 shift toward intronic and other compartments with reduced promoter representation.

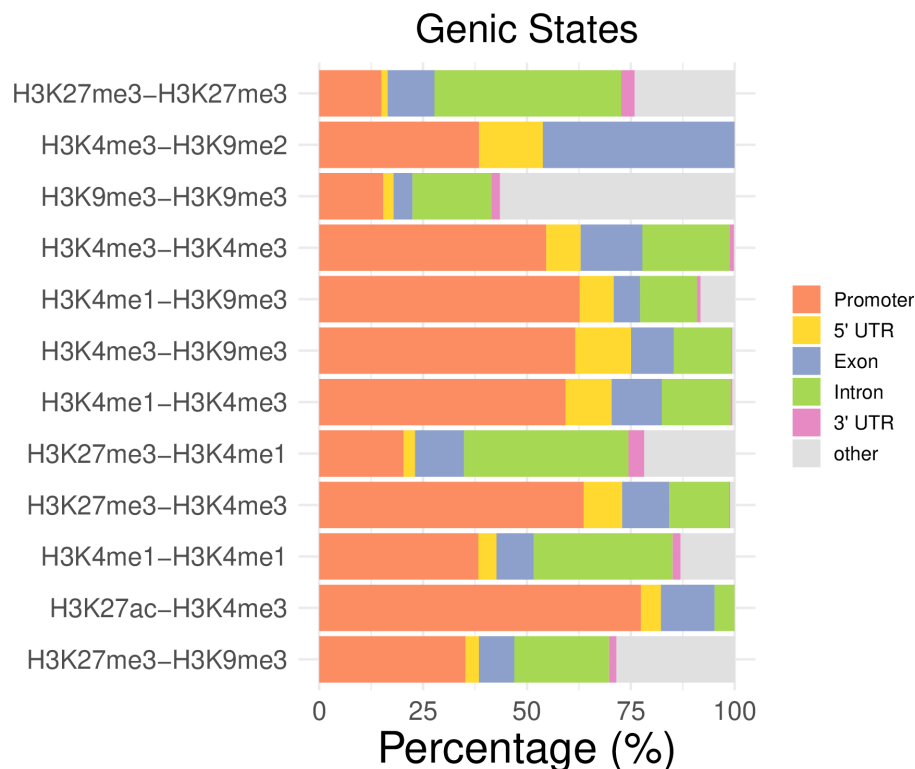


Figure 10. Genic distribution of T2 peaks across histone-mark pairs. Percentages of T2 peaks assigned to promoter, UTR, exon, intron, and other categories are shown for each mark combination. Promoter assignments dominate most pairs (for example H3K4me3–H3K4me3 and H3K27ac–H3K4me3), whereas introns or other categories are more prevalent for H3K27me3–H3K27me3 and H3K9me3–H3K9me3.

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